

Gang Wu*, Xin Wang, Zhishen Wu, Zhiqiang Dong and Qiong Xie

Degradation of basalt FRP bars in alkaline environment

Abstract: This paper investigates the degradation of basalt fiber reinforced polymer (BFRP) bars used for concrete construction in an alkaline environment. The relationships between tensile strength, elastic modulus, shear strength and moisture absorption rate over time are analyzed using a tension test, short-beam test and moisture absorption weighting. The tensile strength degradation of BFRP bars was further compared with that of Glass FRP (GFRP) bars in the literature. The results indicated that BFRP bars exhibit relatively good resistance to alkaline corrosion, maintaining more than 60% of their original strength after 9 weeks at 55°C in an alkaline solution. The moisture absorption of BFRP bars conforms to Fick's law, which shows that the degradation mechanism is controlled by matrix and related interface degradation. This finding is supported by comparison with the shear strength degradation trend. Compared to GFRP bars under similar alkaline conditions, BFRP bars exhibit a similar degradation rate during the initial phase, but maintain higher tensile strength and strength retention over time.

Keywords: alkaline environment; basalt FRP; moisture; shear strength; tensile strength.

*Corresponding author: Gang Wu, International Institute for Urban Systems Engineering, Southeast University, Nanjing, 210096, China, e-mail: g.wu@seu.edu.cn

Xin Wang, Zhiqiang Dong and Qiong Xie: International Institute for Urban Systems Engineering, Southeast University, Nanjing, 210096, China

Zhishen Wu: National and Local Unified Engineering Research Center for Basalt Fiber Production and Application Technology, Southeast University, Nanjing, 210096, China

1 Introduction

Fiber reinforced polymer (FRP) composites, characterized by high strength, light weight and superior corrosion resistance, have been widely accepted as a structural material for repairing existing construction or reinforcing new construction [1–4]. One of the major applications for FRP composites is to replace conventional steel bars in concrete structures [5]. However, due to the increasingly

high cost of carbon FRP (CFRP) and relatively poor resistance to the alkaline of E-glass FRP, the applications of FRPs as structural reinforcement are limited [6].

To incorporate the potential advantages of FRP bars in RC structures, the research has increasingly focused on basalt FRP (BFRP), which demonstrates superior mechanical properties and is more stable chemical resistance compared with E-glass FRP. BFRP also demonstrates a wider range of working temperatures and at a much lower cost than CFRP. Basalt fibers are produced from volcanic rocks and are environmentally friendly, and are nonhazardous materials. However, the degradation of basalt FRP composites under alkaline conditions corresponding to an inner concrete environment requires further evaluation. Many durability studies of FRP composites have focused on the carbon, aramid and glass FRP composites. Only a limited number of studies have been conducted on basalt FRP composites. These reports conclude that the alkaline resistance of basalt fibers is superior to that of glass fibers, whereas the acid resistance is poorer [7, 8]. An investigation of special alkaline-proof basalt fibers showed that their alkaline resistance was superior to their acid resistance, especially in terms of the effect on the flexural modulus [9]. These studies focused on the alkaline resistance of basalt fibers themselves. Because of the various types of basalt fibers and composites that exist, no comprehensive studies have been performed on the degradation behavior and mechanisms of typical BFRP bars in concrete constructions.

In this paper, the structural reinforcement of BFRP bars in an alkaline environment is comprehensively studied including their tensile and shear properties over time. Moisture absorption analysis was conducted to reveal the degradation mechanisms. The alkaline resistance behavior of BFRP and their degradation trend is compared with those of GFRP as reported in the literature.

2 Materials and methods

2.1 Materials and specimens

To comprehensively understand the degradation behavior of BFRP bars under alkaline conditions, the tensile and



Figure 1 Environmental chamber for accelerated corrosion experiment.

shear properties were tested using the accelerated corrosion method. Moisture absorption analysis was also adopted to clarify the degradation trend and mechanism. For each experiment, different specimens were prepared according to the corresponding specifications.

To simulate the concrete environment, the alkaline solution was prepared according to the recommended formula by ACI 440.3R-04 [10]. The formula requires 1 l of water mixed with 118.5 g of $\text{Ca}(\text{OH})_2$, 0.9 g of NaOH and 4.2 g of KOH. All of the specimens were immersed in the solution in a constant temperature environment chamber as shown in Figure 1. The temperatures were set to 25°C and 55°C, respectively, for the experiments according to the accelerated degradation requirement in ACI 440.3R-04. Three aging periods (3 weeks, 6 weeks and 9 weeks) were used to determine the degradation trends of the FRP bars based on previous studies and references in the literature.

2.1.1 Tensile property test

Two types of FRP bars were investigated, including BFRP bars with nominal 6 mm (BFRP-6) and 12 mm diameter (BFRP-12), and a 12 mm diameter steel fiber composite bar (SFCB). The SFCB was produced by longitudinal basalt fibers along a centralized steel bar and can be applied to the concrete members with damage and recoverable ability [11]. Basalt fibers were provided by Zhejiang GBF Basalt Fiber Company, Limited (Hengdian, China) [12] and the FRP bars impregnated with vinyl-ester resin were produced by pultrusion technology. The fiber volume fraction of BFRP bars was approximately 60%, and the proportion of basalt fibers and steel bar was 1–2.71, in which the basalt fibers was composed of 16 bundles of 4800 tex fiber roving and the corresponding BFRP layer occupied a thickness of



Figure 2 Surface condition and cross-section of FRP bars.

1 mm. The surfaces of all FRP bars were ribbed by a nylon laminate during pultrusion, as shown in Figure 2.

The specimens for the tensile properties test were developed according to the JSCE-E 531 guidelines [13] and are shown in Figure 3. The both ends were treated by sand blasting over a length of 250 mm and anchored with seamless steel tubes 2 mm thick with outer diameters of 14 mm and 20 mm for BFRP-6 and BFRP-12 bars, respectively. An epoxy resin was used to fill the gap between the steel tube and the FRP bar, and the specimen was cured for 7 days to ensure sufficient strength. Five specimens were tested for each condition. A total number of 105 specimens were tested and 90 of these specimens were used for long-term durability testing. The number of tensile specimens in alkaline solutions is listed in Table 1.

2.1.2 Shear property test

The shear properties of the FRP bar in an alkaline solution were tested by the short-beam testing method according to ASTM 4475-02 [14]. Because short-beam testing primarily determines the interlaminar strength of FRP, the shear

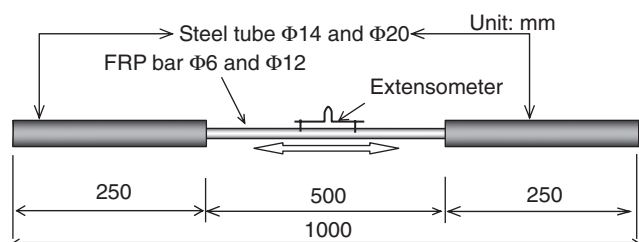


Figure 3 Specimens preparation for tensile test.

Table 1 Tensile specimen arrangement.

Temperature	25°C			55°C		
Aging time (week)	3	6	9	3	6	9
Type of FRP bar						
BFRP-6	5	5	5	5	5	5
BFRP-12	5	5	5	5	5	5
SFCB	5	5	5	5	5	5

strength of FRP bars cannot be used for shear design purposes. However, the results can be referred to in order to compare the variations in shear properties after corrosion. The specimens of BFRP bars with a 12 mm diameter were used. The total length of each specimen was 48 mm, and 5 specimens were prepared for each test condition. The specimens were immersed in alkaline solution in the environmental chamber. A total of 20 specimens was used to determine the original shear strength and shear strength degradation after 3, 6, and 9 weeks in an alkaline solution.

2.1.3 Moisture absorption test

The moisture absorption test was conducted using the BFRP bars with a diameter of 12 mm. The length of each specimen was 100 mm, and three specimens were prepared for each group. A total of 30 specimens was used to determine the moisture diffusion overtime (Table 2).

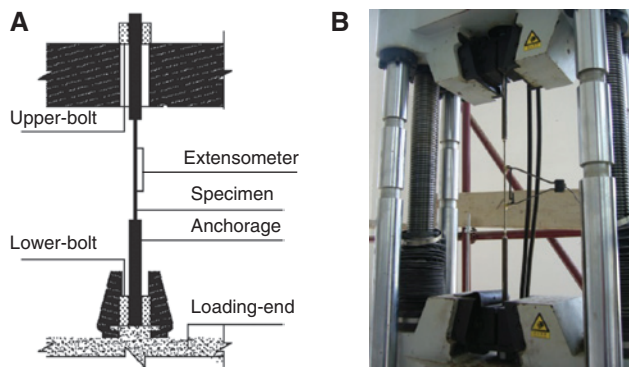
2.2 Test set-up and procedures

The tensile property test is conducted using a universal MTS tension-compression test machine with a capacity of 500 kN (Suns, Shenzhen, China). The specimens, with the anchors at both ends, was affixed by clamps, as shown in Figure 4. The loading rate was at 2 mm/min until failure.

The shear capacity test and the short-beam test were conducted using the same MTS tension-compression test machine. A two-point support was set up at the bottom-loading end. The 48 mm long specimen was placed in the supports with a panning of 36 mm to ensure inter-laminate shear failure (Figure 5). The specimens were loaded at a rate of 2 mm/min up to failure.

Table 2 Specimens arrangement for moisture absorption test.

Aging time (day)	0	1	2	7	14	21	28	35	49	63
Number of specimens	3	3	3	3	3	3	3	3	3	3

**Figure 4** Tensile test set-up: (A) Schematic configuration (B) specimens in the machine.

The shear strength was calculated by the following equation recommended by ASTM.

$$\tau_{\max} = \frac{V}{3I_z} \left(\frac{d}{2} \right)^2 = \frac{8}{3} \frac{F}{\pi d^2} \quad (1)$$

Where F is the load at shear failure and d is the diameter of the BFRP bar.

For the moisture absorption test, the specimens were removed from the solution at their respective aging times. The solution attached on the surface was removed with alcohol and the surface was dried with a cotton tissue. The specimen was weighed (w) using an electronic balance with precision of 0.0001 g. The samples underwent further drying at a constant temperature of 60°C by an electro-thermostatic blast oven (Figure 6) until the weight was constant. The specimens were weighed again (w_d). The moisture absorption rate (M) was calculated by the following equation.

$$M = \frac{\Delta w}{w_d} \times 100\% = \frac{w - w_d}{w_d} \times 100\% \quad (2)$$

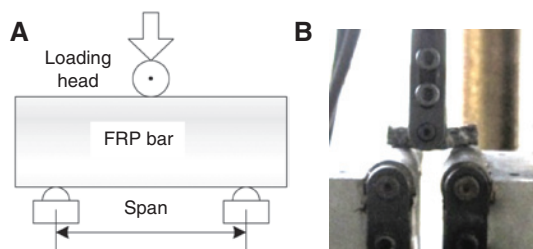
**Figure 5** Specimen and loading device for shear property test: (A) Schematic configuration (B) specimens in the machine.



Figure 6 Electro-thermostatic blast oven.

3 Results and discussion

3.1 Surface condition

The surface condition changes after 3 weeks of immersion in the alkaline solution. The etching in the surface was observed, and a jelly-like substance was seen on the surface. After drying, the surface of the corroded BFRP bars were no longer smooth, and partial fibers were found to be dry and loose. The jelly was probably generated by the reaction between the resin and corrosive elements. At 55°C, the surface showed more etching, unsticking of fibers and resin compared with

degradation at 25°C. The corroded surface conditions are shown in Figure 7.

3.2 Tensile property degradation

3.2.1 Experimental results

The tensile failure modes of BFRP bars were constant regardless of the time taken. All of the BFRP specimens failed in the middle portion and exhibited dispersion as shown in Figure 8 (A,B). The failure mode of SFCBs was slightly different compared with the BFRP bars, which exhibited fracture of fibers in the middle portion, but maintained a bundled FRP configuration without fiber dispersion. The failure mode of SFCB indicated that the steel bar inside the BFRP prevented sudden failure.

The degradation tensile properties of the three types of FRP bars in an alkaline solution are shown in Table 3.

The results are plotted in Figures 9 and 10 to show the differences and trend in tensile property degradation overtime.

3.2.2 Discussion

Figure 9 depicts the degradation tensile strength for different types of FRP bars with respect to time. The degradation rate was highest in the first few weeks of corrosion and then slowed SFCB under alkaline conditions at 25°C, except for that of BFRP-12. This result indicates that at a lower temperature, the corrosive elements cannot permeate the thicker FRP bars as easily. Although the degradation rates of BFRP-6 specimens at 55°C were larger than those of the other specimens, the residual strength remained the highest among other specimens under identical conditions due to their original strengths. This finding suggested restriction of this material under certain



Figure 7 Surface degradation of corroded BFRP bars: (A) original surface (B) after 6 weeks under 25°C alkaline solution (C) after 6 weeks under 55°C alkaline solution.



Figure 8 Failure modes of different FRP bars: (A) BFRP-6 (B) BFRP-12 (C) SFCB.

environmental design conditions. Due to the combination of basalt fibers and an alkaline-resistant steel bar, the SFCBs demonstrated a relatively small degradation, suggesting its applicability in concrete structures.

Figure 10 demonstrates that the elastic modulus of different FRP bars remained almost constant in the observed aging over time. The variations in amplitudes of the elastic modulus were limited, below 4% for BFRP-6 and BFRP-12 bars, and <10% for SFCB bars. No relationship between the variation of the elastic modulus and corrosion time was observed.

The coefficients of variation (CVs) of tensile strength for BFRP bars were relatively larger than that of the SFCB after corrosion regardless of the temperature (Table 3). The CV of tensile strength of BFRP bars also increased

overtime, whereas the CV of tensile strength of the SFCB did not increase over time. This phenomenon indicated that the corroded BFRP bars exhibited more scatter due to their elastic and brittle behavior, and the elasto-plastic and ductile steel bars inside SFCB could have benefited from the stability of strength after corrosion.

The CV of modulus of SFCBs was generally larger than that of BFRP bars. The variation of the modulus primarily reflected the continuous interface degradation of fibers and resin in the FRP. This result indicated that the interface between basalt fibers and steel bar of SFCB was relatively weaker than the interfaces of fibers and resin in BFRP bars. However, it could only have affected the scatter of the modulus after corrosion but did not affect strength due to the ductility of the steel bar inside.

Table 3 Degradation of tensile property of FRP bars.

Type of FRP	Temperature	Aging time (week)	Tensile strength			Elastic modulus		
			Mean (MPa)	CV (%)	Retention (%)	Mean (MPa)	CV (%)	Retention (%)
BFRP-6	25°C	Control	1325	0.71	100.00	56	2.16	100.00
		3	1021	0.45	77.10	55	0.31	98.36
		6	922	3.18	69.64	55	2.66	97.42
		9	919	7.84	69.38	55	0.45	97.77
	55°C	3	885	3.71	66.81	56	0.77	98.57
		6	818	4.47	61.72	54	1.93	96.40
		9	802	11.36	60.58	55	1.07	97.66
	Control	3	1089	5.66	100.00	46	3.84	100.00
		6	1036	15.21	95.18	48	2.86	103.73
BFRP-12	25°C	3	910	1.02	83.58	48	2.49	104.52
		6	893	6.22	82.03	47	4.62	103.12
		9	807	11.79	74.11	47	3.50	101.71
	55°C	3	807	11.79	74.11	47	3.50	101.71
		6	681	15.76	62.55	47	3.14	103.05
		9	669	16.53	61.40	45	4.37	98.43
	Control	3	577	1.23	100.00	155	3.12	100.00
		6	565	2.88	97.97	147	8.62	94.73
		9	507	3.47	87.97	156	14.10	100.56
SFCB	25°C	3	488	1.70	84.67	140	2.06	90.34
		6	453	3.31	78.56	149	6.19	95.71
		9	414	2.25	71.82	156	14.85	100.50
	55°C	3	406	2.47	70.42	157	8.76	101.30
		6						
		9						
	Control	3						
		6						
		9						

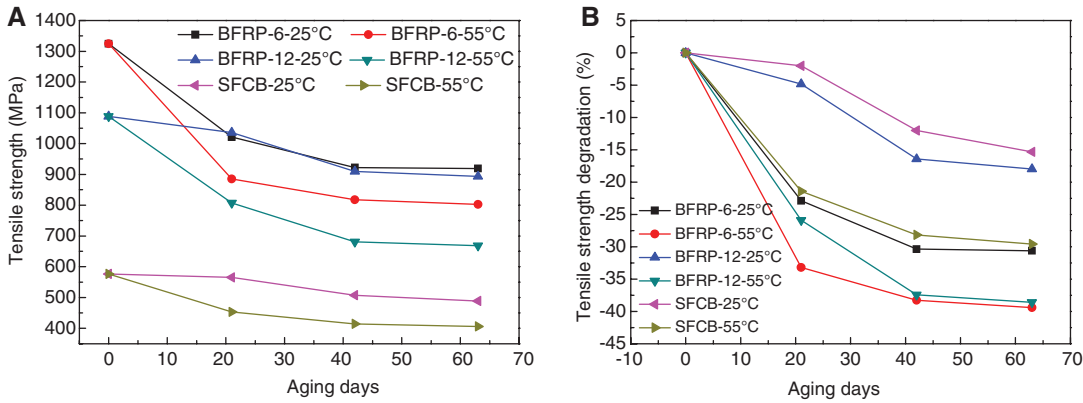


Figure 9 Tensile strength degradation of different FRP bars: (A) tensile strength degradation (B) tensile strength retention.

The degradation of interfaces inside the FRP plays an important role to the degradation of the overall strength, and the resin could have effectively protected fibers from corrosion by alkaline solutions.

3.3 Shear property degradation

The degradation of the shear property of BFRP-12 FRP bars at 55°C in an alkaline solution is shown in Table 4 and Figure 11. Over time, the shear strength degraded; however, differences in the strength degradation over 3, 6, and 9 weeks were not observed. The degradation of shear capacity was almost at a maximum within the first 3 weeks. The shear strength retention of the BFRP bars was much higher than the tensile strength retention under identical conditions because the shear capacity test/short-beam test primarily reflected the interlaminar shear resistance of FRP. The lowering of interlaminar shear strength demonstrated that the degradation occurred on the interfaces

between fibers and resin inside the FRP, which resulted in a decrease of tensile strength. The degradation rate of shear strength (15%) was much lower than that of tensile strength (38.6%) after 9 weeks of immersion in the 55°C alkaline solution. The interlaminar shear strength was influenced strongly by the middle layer in which the interlaminar shear stress was at its maximum, whereas the tensile strength was governed by the force resistance of the full cross-section. Therefore, the middle layer suffered from less corrosion and the degradation amplitude of shear strength and the severely corroded outer layer of the FRP bar accelerates the tensile strength degradation.

3.4 Moisture absorption test

Moisture absorption mainly reflected the diffusion of corrosive elements in the FRP, which could have been modeled by Fick's law. Fick's law derived by Adolf Fick in 1855, is an observed law stating that the rate at which

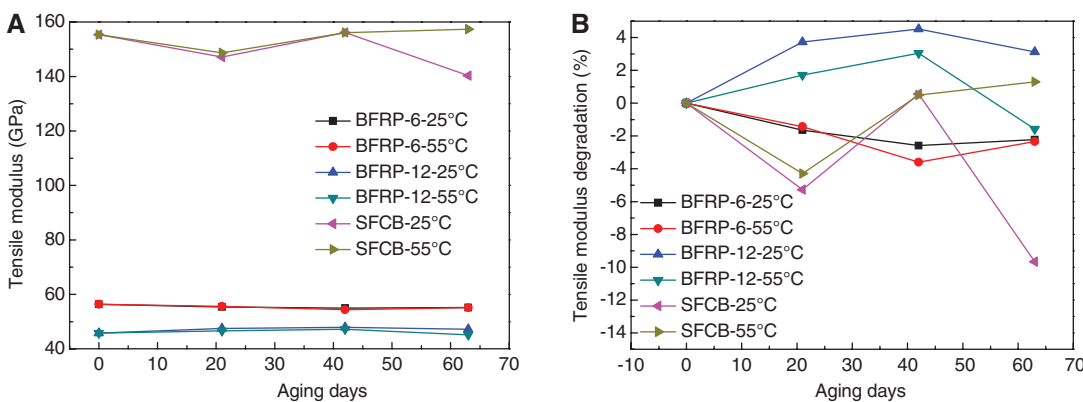


Figure 10 Elastic modulus degradation of different FRP bars: (A) elastic modulus degradation (B) elastic modulus retention.

Table 4 Shear strength degradation of BFRP-12.

Aging time (week)	Mean (MPa)	SD (MPa)	CV (%)	Retention (%)
0	22.34	0.89	3.98	100
3	19.50	0.23	1.18	87.31
6	19.37	0.46	2.37	86.72
9	18.97	0.78	4.11	84.95

Note: SD means standard deviation, CV means coefficient of variation.

one substance diffuses through another is directly proportional to the concentration gradient of the diffusing substance. Some models based on Fick's law were developed to describe the diffusion process in the FRP bars [15, 16]. These models were used to determine the diffusion coefficient from the plots of moisture content versus square root of time as shown in Figure 12A.

In the phase of Fickian diffusion, the corrosive ions diffuse according to Fick's law. After this stage, due to the reaction of OH^- ions and matrix of FRP bar, the penetration of OH^- will cause micro-cracking and fracturing of the matrix and result in rapid moisture diffusion as well as debonding of the fibers. That process describes the non-Fickian phase of moisture absorption. The following moisture absorption of BFRP bars, with respect to the square root of time, further demonstrated the relationship above.

The moisture absorption rates of BFRP-12 bars at 55°C relationship in an alkaline solution are listed in Table 5. The relationship between the moisture absorption rate and corrosion time is triphasic (Figure 12B). In the first phase (the initial 3 weeks), the moisture was rapidly absorbed by the BFRP bars. Then the moisture absorption enters a second phase with a steady increase up to 8 weeks. In the last phase, the absorption rate increased

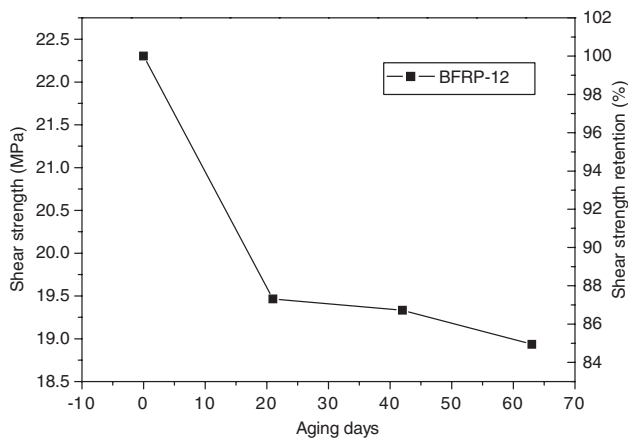
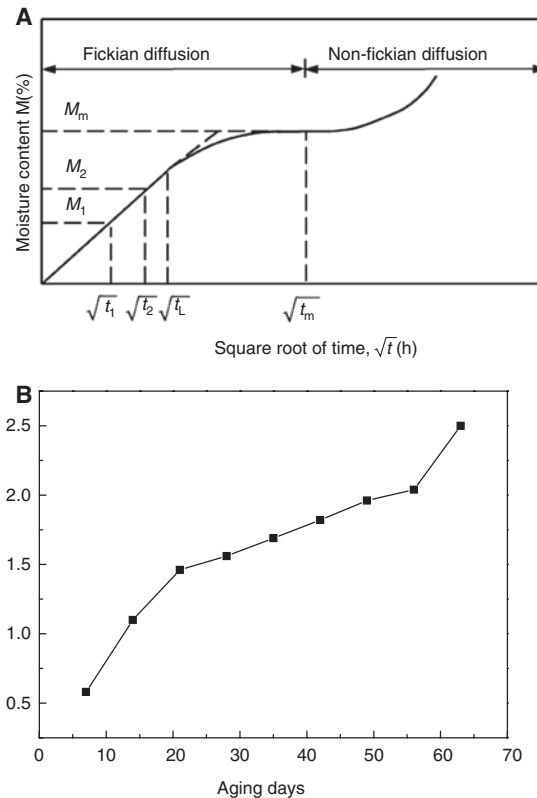
**Figure 11** Relationship of shear strength and aging time.

Figure 12 Relationship between moisture absorption and aging days: (A) Typical relationship between moisture absorption and square root of time [17]. (B) Tested relationship between moisture absorption and aging days.

again. The first two phases conform to Fick's law, whereas the last phase represents non-Fickian diffusion. In the last phase, the rapid increase of moisture content may have been caused by the micro cracks of the matrix in the FRP bar generated by the excessive moisture content over time.

It is also observed that the CV of moisture content during the first few weeks of the second phase exhibited a relatively higher level compared to those in the first and last phases (Table 5). This result suggested that in the first phase, due to the unsaturation status, the moisture absorption maintains at an even rate so that the CV exhibited a low level. With the increase of moisture diffusion, the FRP became saturated in the solution. In this phase, the moisture diffusion was primarily affected by random inner defects such as voids. Thus, the scatter of the moisture absorption rate became larger. When diffusion was in the final phase, the scattering of moisture content was decreased again due to the contribution the uniform matrix cracking, which resulted because the FRP was fully saturated.

Table 5 Moisture absorption rate.

Aging period (week)	Mean (%)	SD (%)	CV (%)
1	0.58	0.02	3.45
2	1.10	0.10	9.09
3	1.46	0.10	6.85
4	1.56	0.24	15.38
5	1.69	0.17	10.06
6	1.82	0.35	19.23
7	1.96	0.12	6.12
8	2.04	0.10	4.90
9	2.50	0.03	1.20

Because moisture absorption reflected the degradation of the matrix and its interfaces with fibers, the moisture-developing trend could have also been compared with shear strength degradation. The rapid drop of shear strength in the first three weeks corresponded to the fast absorption of moisture in the first phase. The final decrease of shear strength after 9 weeks also corresponded to the third phase of moisture diffusion. This similarity indicated that the increase of moisture in the matrix and their interfaces with the fibers degrades simultaneously, resulting in the loss of shear and tensile strength. The effective prevention or delay of moisture diffusion in the matrix could have potentially enhanced the corrosion resistance of FRP.

4 Comparison analysis

The mechanical properties of basalt fibers and glass fibers are often compared due to their similar chemical compositions [18]. In general, basalt fiber composites exhibit approximately 20% higher tensile strength and modulus than E-glass fiber composites [19]. It is also reported that basalt fibers show relatively more chemical resistance and wider working temperatures compared with E-glass fibers. In this study, the tensile strength degradation of basalt FRP bars was compared with the degradation of glass FRP bars under similar alkaline conditions. The parameters for the current study were selected in order to closely model the existing literature [20–22]. However, due to the wide variability in reported conditions, our research only discussed the trends in strength degradation between BFRP and GFRP. The results are shown in Figure 13.

Figure 13A shows that the initial tensile strength of the BFRP bar is 1.43 and 3.66 times than that of E-glass/vinylester FRP and E-glass/polyester FRP bars,

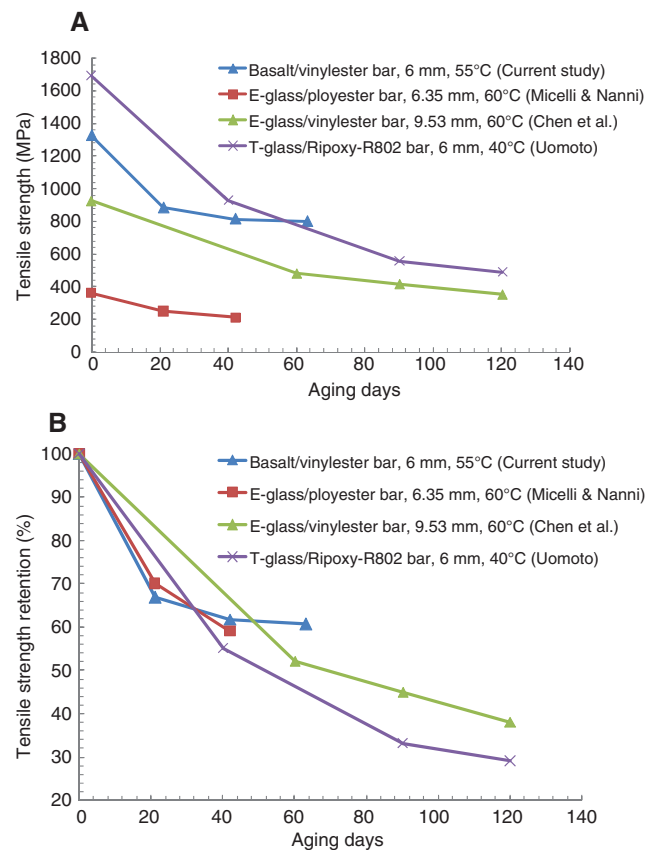


Figure 13 Comparison of tensile strength of BFRP and GFRP under alkaline solution: (A) tensile strength degradation (B) tensile strength retention.

respectively. The low strength of the E-glass/polyester FRP bars was the result of the impregnation with polyester resin. T-glass/ripoxyl FRP bars exhibit the highest initial strength. However, the residual strength of T-glass/ripoxyl FRP bars was lower than that of BFRP bars after 63 days corrosion. Figure 13B shows the degradation rates for different BFRP and GFRP bars. Although E-glass/vinylester FRP bars exhibited the lowest degradation rate in the first few weeks compared with other GFRP and BFRP bars, the strength retention maintains a decreasing trend and was <40% of the initial strength after 120 days of corrosion. Although BFRP bars degrade fast initially, the degradation rate became slow after the first 3 weeks. The corresponding strength retention maintains 61% of its initial strength after 63 days of corrosion. In the first few weeks of corrosion, no obvious difference of strength degradation was observed for BFRP and GFRP bars. As time increased, the BFRP bar exhibits a steady and slow rate of degradation, whereas the GFRP bar continued rapid degradation. It can be concluded that BFRP potentially has superior alkaline resistance to E-glass FRP. However,

due to the relatively short aging time and insufficient data from previous research, it was still difficult to strictly evaluate their behavior. Based on the comparison above, it was possible to show application of BFRP bars in concrete structures according to the experiences and guidelines of GFRP bars in a similar environment.

5 Conclusions

This paper investigated the mechanical property degradation of BFRP and SFCB bars in an alkaline solution by means of accelerated corrosion test. The degradation of tensile and shear properties as well as the moisture absorption rate were studied. The characteristics of degradation were analyzed and summarized. The experimental results were compared with the degradation of GFRP reported in the literature. The major conclusions are as follows:

1. BFRP bars exhibit relatively good resistance to alkaline corrosion maintaining more than 60% of their original strength after 9 weeks in a 55°C alkaline solution. The small diameter BFRP bars demonstrated a similar degradation rate but maintained higher residual strength compared with the BFRP bars with a larger diameter. SFCB bars display stronger resistance to alkaline corrosion due to the steel bar inside.
2. Shear strength degradation was similar to tensile strength degradation. The relatively small degradation amplitude of shear strength was induced by a different failure mechanism between tensile and shear strength.
3. The moisture absorption trend of BFRP bars conformed with Fick's law and demonstrates the degradation mechanism controlled by matrix and related interfaces degradation. This finding was further verified by comparison with the shear strength degradation trend.
4. BFRP bars exhibited a similar degradation rate in the initial phase of corrosion and maintain higher tensile strength and strength retention compared with GFRP bars in similar alkaline corrosive solutions. A slow rate of degradation was observed for BFRP after the initial phase of degradation.

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